Assignment 2 - AIML Lab (B4)

1. Write a program that asks the user for a positive integer value. The program should calculate the sum of all the integers from 1 up to the number entered. For example, if the user enters 20, the loop will find the sum of 1, 2, 3, 4, ... 20.

```
n = int(input("Enter a positive integer: "))

sum = 0

for i in range(1, n + 1):
    sum += i

print("Sum =", sum)
```

Output:

```
Enter a positive integer: 20
Sum = 210
```

2. Write a program that prompts the user to input a number and prints its factorial. The factorial of an integer n is defined as n! = 1 x 2 x 3 x ... x n; if n > 0 = 1; if n = 0 For instance, 6! can be calculated as 1 x 2 x 3 x 4 x 5 x 6. [REPEATED]

```
n = int(input("Enter a number: "))

fact = 1

for i in range(1, n + 1):
    fact = fact * i

print("Factorial =", fact)
```

Output:

```
Enter a number: 6
Factorial = 720
```

3. Write a program that prompts the user to input a number and reverse its digits. For example, the reverse of 12345 is 54321; reverse of 5600 is 65

```
n = int(input("Enter a number: "))

reverse = 0

while n > 0:
    digit = n % 10
    reverse = reverse * 10 + digit
    n = n // 10

print("Reverse =", reverse)
```

Output 1:
Enter a number: 12345
Reverse = 54321

Output 2:
Enter a number: 5600
Reverse = 65

4. Write a program that asks the user to input a positive integer. Your program should find and display the sum of digits of number. For example, sum of digits of number 32518 is 3+2+5+1+8 = 19.

```
n = int(input("Enter a positive integer: "))

sum = 0

while n > 0:
    digit = n % 10
    sum += digit
    n = n // 10

print("Sum of digits =", sum)
```

Output:
Enter a positive integer: 32518
Sum of digits = 19

5. A palindromic number is a number that remains the same when its digits are reversed. For example, 16461. Write a program that prompts the user to input a number and determine whether the number is palindrome or not.

```
n = int(input("Enter a number: "))

original = n
reverse = 0

while n > 0:
    digit = n % 10
    reverse = reverse * 10 + digit
    n = n // 10

if original == reverse:
    print("Palindrome")
else:
    print("Not Palindrome")
```

Output:

```
Enter a number: 16461
Palindrome
```

6. Write a program to enter the numbers till the user wants and at the end it should display the count of positive, negative and zeros entered.

```
positive = 0
negative = 0
zero = 0

while True:
    n = int(input("Enter a number: "))

    if n > 0:
        positive += 1
    elif n < 0:
        negative += 1
    else:
        zero += 1

    choice = input("Continue? (y/n): ")
    if choice == 'n':
        break

print("Positive =", positive)
print("Negative =", negative)
print("Zero =", zero)
```

Output:

```
Enter a number: 10
Continue? (y/n): y
Enter a number: -5
Continue? (y/n): y
Enter a number: 0
Continue? (y/n): y
Enter a number: 8
Continue? (y/n): n

Positive = 2
Negative = 1
Zero = 1
```

7. Write a program to enter the numbers till the user wants and at the end the program should display the largest and smallest numbers entered.

```
largest = None
smallest = None
```

```
while True:
    n = int(input("Enter a number: "))

    if largest is None or n > largest:
        largest = n

    if smallest is None or n < smallest:
        smallest = n

    choice = input("Continue? (y/n): ")
    if choice == 'n':
        break

print("Largest =", largest)
print("Smallest =", smallest)
```

Output:

```
Enter a number: 10
Continue? (y/n): y
Enter a number: 25
Continue? (y/n): y
Enter a number: 5
Continue? (y/n): y
Enter a number: 15
Continue? (y/n): n

Largest = 25
Smallest = 5
```

8. Write a program that accepts a string from user. Your program should count and display number of vowels in that string.

```
text = input("Enter a string: ")

count = 0

for ch in text:
    if ch in "aeiouAEIOU":
        count += 1

print("Number of vowels =", count)
```

Output:

```
Enter a string: Hello World
Number of vowels = 3
```

9. Write a Python program that accepts a string from user. Your program should create and display a new string where the first and last characters have been exchanged.

```
text = input("Enter a string: ")

new_string = text[-1] + text[1:-1] + text[0]

print("New string =", new_string)
```

Output:

Enter a string: Python
New string = nythoP

10. Write a Python program that accepts a string from the user. Your program should create a new string that is the reverse of the given string and display it.

```
text = input("Enter a string: ")

reverse = text[::-1]

print("Reverse =", reverse)
```

Output:

Enter a string: Python
Reverse = nohtyP

11. Write a program that asks the user to input his name and print its initials. Assuming that the user always types first name, middle name and last name and does not include any unnecessary spaces.
For example, if the user enters Ajay Kumar Garg the program should display A. K. G.

```
name = input("Enter your name: ")

words = name.split()

for word in words:
   print(word[0] + ".", end=" ")
```

Output:

Enter your name: Ajay Kumar Garg
A. K. G.

12. Write a program in python that accepts a string to setup a passwords. Your entered password must meet the following requirements:

- **The password must be at least eight characters long.**
- **It must contain at least one uppercase letter.**
- **It must contain at least one lowercase letter.**

- **It must contain at least one numeric digit.**

Your program should should perform this validation.

```
password = input("Enter password: ")

if (len(password) >= 8 and
    any(c.isupper() for c in password) and
    any(c.islower() for c in password) and
    any(c.isdigit() for c in password)):

    print("Valid Password")
else:
    print("Invalid Password")
```

Output:

```
Enter password: Hello123
Valid Password
```

13.Write a Python program that accepts an n × n matrix from the user. Calculate the sum of each row and display the matrix rows in ascending order of their row sums. [EXTRA]

Example:

Input matrix:

1 2 3
9 1 2
4 1 1

Row sums:

- **1 + 2 + 3 = 6**
- **9 + 1 + 2 = 12**
- **4 + 1 + 1 = 6**

Sorted matrix:

1 2 3
4 1 1
9 1 2

```
n = int(input("Enter n: "))

matrix = []

for i in range(n):
    row = list(map(int, input("Enter row: ").split()))
    matrix.append(row)

matrix.sort(key=sum)

print("Sorted matrix:")

for row in matrix:
    print(row)
```

Output:

```
Enter n: 3
```

Enter row: 1 2 3
Enter row: 9 1 2
Enter row: 4 1 1
Sorted matrix:
[1, 2, 3]
[4, 1, 1]
[9, 1, 2]